



Monday Feb-24-2025

Flowserve Facility:

Flowserve Nordstrom / Valbart Valves
1511 Jefferson St.
Sulphur Springs, TX 75482

cmtrs@flowserve.com

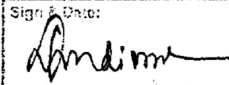
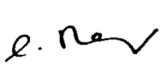
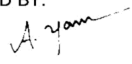
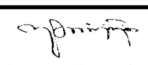
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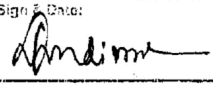
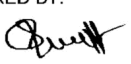
CERTIFICATE OF CONFORMANCE

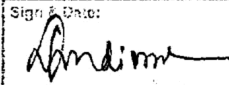
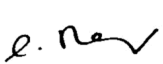
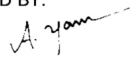
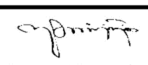
SERIAL #	HEAT #	PO #	Description	Class	End Connection
61-691957	1N415	1173982	4" 2345	900	FLG
61-691958	0299B	1173982	4" 2345	900	FLG
61-691956	1N416	1173982	4" 2345	900	FLG
61-691955	1N418	1173982	4" 2345	900	FLG
61-691707	1N420	1173344	4" 2345	900	FLG
61-691710	4026	1173344	4" 2345	900	FLG
61-691709	1N415	1173344	4" 2345	900	FLG
61-691708	1N415	1173344	4" 2345	900	FLG
61-691706	4348	1173344	4" 2345	900	FLG
61-691705	4363	1173344	4" 2345	900	FLG

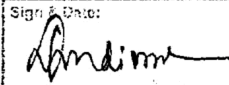
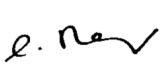
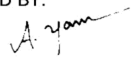
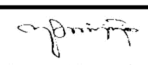
We hereby certify that all the subject valves/parts/sealant have been manufactured in conformance with the applicable requirements of Flowserve Sulphur Springs Operation (FSSO) specifications and that the valve/parts/sealant materials for component parts meet the applicable requirements of FSSO drawings and ASTM specifications, ASME B16.34 (If Applicable), API Q1 10th edition, and API 6D 25th edition.(if applicable to the API monogram). Flowserve Dynamic Balanced Valves are provided with a Pressure balance feature unless specifically identified in the Valve description. All valves/parts/sealant meet all the requirements of the customer purchase order.

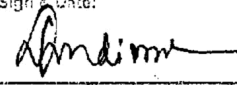
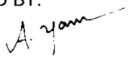
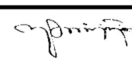
Documentation Technician
Justin McKee

	FLOW LINK SYSTEMS PRIVATE LIMITED FOUNDRY UNIT-III-SF NO 395/2B2, KEMPANAICKANPALAYAM, ANNUR, COIMBATORE - 641 653, INDIA		INSPECTION CERTIFICATE (EN 10204:2004-3.1)		TC NO : 0816 DATE : 28-Jun-2024									
	CUSTOMER FLOWSERVE INDIA CONTROLS PVT LTD, B8, MMDA Industrial Area, Maraimalai Nagar-603209 Kancheepuram, INDIA		SPECIFICATION REFERENCE 1. ASTM A216-21 Gr.WCC 2. Customer Spec. RMC 90012 Rev. 15 3. NACE MR0175-2021 / ISO15156-2020		HEAT NO. 1N415 Date of Pouring 27-Mar-2024 FOUNDRY MARK FLS									
PO NUMBER & DATE 10115103 Dt : 24-Nov-2023			ORDER ACCEPTANCE NO. & DATE SO-24-0377											
CHEMICAL COMPOSITION % wt														
MELTING PRACTICE : Steel melted using Electric Induction furnace, and fully killed using Aluminium.														
SPECIFICATION	C	Si	S	P	Mn	Ni	Cr	Mo	V	Cu	Al	--	*RE	**CE
Minimum	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum	0.23	0.60	0.020	0.025	1.20	0.50	0.50	0.20	0.030	0.30	0.08	--	1.00	0.43
Achieved	0.194	0.466	0.009	0.015	0.979	0.058	0.087	0.017	0.005	0.011	0.041	--	0.18	0.38
Heat Treatment : NORMALISING: Temperature raised to 920°C, Soaked for 3 hours and then atmospheric air cooled														
Visual Inspection - Castings conforms to ANSI/MSS-SP-55-2011 - Type II to XII (a)														
MECHANICAL PROPERTIES														
TEST TYPE	Tensile as per ASTM A370-22 / ISO 6892-1-2020 (at Room Temperature)						Impact (CVN) ISO 148-2016/ASTM A370-22		BEND TEST		HARDNESS			
SPECIFICATION	YS Rp 0.2% (KSI)	YS Rp 1.0% (KSI)	UTS (KSI)	% Elongation 4d 5d		% RA	Test Temperature (at °C) J Values Avg.		ANGLE	D	(HBW / HRC)			
Minimum	40	--	70	22	--	35	--	--	--	--	--			
Maximum	--	--	95	--	--	--	--	--	--	--	237			
Achieved	45.8	--	77.0	36.0	--	60.0	--	--	--	--	154			
DETAIL OF CASTINGS														
S.No.	PO S.No.	DESCRIPTION	CAST DRAWING NO / REV	PART NO		POURED QTY	DESP. QTY							
1	01	4" 900 FE BODY	D-487876 / 20	N2757928-60		5	1							
REMARKS: 1. Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast keel block. 2. *RE = Ni + Cr + Mo + V + Cu 3. ** CE = C + $\frac{Mn}{6}$ + $\frac{Cr+Mo+V}{5}$ + $\frac{Ni+Cu}{15}$ 4. The above stated KSI values have been calculated based on original tested N/mm ² values. 5. Stated actual Hardness value is less than 22 HRC. 6. Degassing process conform as per RMP 61496 Rev A.														
CERTIFIED THAT 1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO. 2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1. 3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIO ACTIVE ELEMENT OR RADIATION.				INSPECTED / REVIEWED & ACCEPTED / NOT ACCEPTED Sign & Date:  Sample Qty: M. VADIVEL - FICPI, QA  C. RAJAN ASSISTANT MANAGER - QUALITY MANUFACTURER'S AUTHORIZED INSPECTOR										
PREPARED BY: A.Yamuna 		CHECKED BY: N.KARTHICK  QUALITY ASSURANCE ENGINEER												

	FLOW LINK SYSTEMS PRIVATE LIMITED FOUNDRY UNIT - II 65 / 2-A,B & C,PUDUPALAYAM B.P.O AVINASHI TK, COIMBATORE - 641 654 INDIA	INSPECTION CERTIFICATE (EN 10204:2004-3.1)	TC NO : A-0833 DATE : 10-May-2024											
CUSTOMER FLOWSERVE INDIA CONTROLS PVT LTD, B8, MMDA Industrial Area, Maraimalai Nagar-603209 Kancheepuram, INDIA		SPECIFICATION REFERENCE 1. ASTM A216-21 Gr.WCC 2. Customer Spec. RMC 90012 Rev. 15 3. NACE MR 0175-2021 / ISO 15156 - 2020	HEAT NO. 0299B Date of Pouring 06-Mar-2024 FOUNDRY MARK FLS											
PO NUMBER & DATE 10115280 Dt. 05.12.2023		ORDER ACCEPTANCE NO 24-0223												
CHEMICAL COMPOSITION % wt														
MELTING PRACTICE : Steel melted using Electric Induction furnace, and fully killed using Aluminium.														
SPECIFICATION	C	Si	S	P	Mn	Ni	Cr	Mo	V	Cu	Al	--	*RE	**CE
Minimum	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum	0.23	0.60	0.020	0.025	1.20	0.50	0.50	0.20	0.030	0.30	0.08	--	1.00	0.43
Achieved	0.189	0.414	0.008	0.013	0.923	0.056	0.109	0.015	0.004	0.018	0.041	--	0.20	0.37
Heat Treatment : NORMALIZING: Temperature raised to 920°C, Soaked for 4 hours and then atmospheric air-cooled														
Visual Inspection - Castings conforms to ANSI/MSS-SP-55-2011 - Type II to XII (a)														
MECHANICAL PROPERTIES														
TEST TYPE	Tensile as per ASTM A370-22 / ISO 6892-1-2020 (at Room Temperature)						Impact (CVN) ISO 148-2016/ASTM A370-22		BEND TEST		HARDNESS			
SPECIFICATION	YS Rp 0.2% (KSI)	YS Rp 1.0% (KSI)	UTS (KSI)	% Elongation 4d 5d		% RA	Test Temperature (at °C) J Values Avg.		ANGLE	D	(HBW / HRBW)			
Minimum	40	--	70	22	--	35	--	--	--	--	--			
Maximum	--	--	95	--	--	--	--	--	--	--	237			
Achieved	45.3	--	75.1	34.8	--	64.5	--	--	--	--	152			
DETAIL OF CASTINGS														
S.No.	PO S.No.	DESCRIPTION	CAST DRAWING NO / REV	PART NO	POURED QTY	DESP. QTY								
1	03	4" 900 FE BODY	D-487876 / 20	N2757928-60	5	3								
REMARKS: 1. Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast keel block. 2. *RE = Ni + Cr + Mo +V +Cu $** CE = C + \frac{Mn}{6} + \frac{Cr+Mo+V}{5} + \frac{Ni+Cu}{15}$ 3 4. The above stated KSI values have been calculated based on original tested N/mm ² values. 5. Stated actual Hardness value is less than 22 HRC. 6. Degassing process conform as per RMP 61496 Rev. A														
INSPECTED / REVIEWED & ACCEPTED / NOT ACCEPTED Sign & Date:  Sample Qty: M. VADIVEL - FICPI, QA														
CERTIFIED THAT 1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO. 2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1. 3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIO ACTIVE ELEMENT OR RADIATION.														
PREPARED BY:  S.KARTHICK			CHECKED BY: A.LOGANATHAN LAB CHEMIST 			 M.S.DHANABALAN DEPUTY MANAGER - QUALITY MANUFACTURER'S AUTHORIZED INSPECTOR								

	FLOW LINK SYSTEMS PRIVATE LIMITED FOUNDRY UNIT-III-SF NO 395/2B2, KEMPANAICKANPALAYAM, ANNUR, COIMBATORE - 641 653, INDIA		INSPECTION CERTIFICATE (EN 10204:2004-3.1)		TC NO : 0817 DATE : 28-Jun-2024									
	CUSTOMER FLOWSERVE INDIA CONTROLS PVT LTD, B8, MMDA Industrial Area, Maraimalai Nagar-603209 Kancheepuram, INDIA		SPECIFICATION REFERENCE 1. ASTM A216-21 Gr.WCC 2. Customer Spec. RMC 90012 Rev. 15 3. NACE MR0175-2021 / ISO15156-2020		HEAT NO. 1N416 Date of Pouring 27-Mar-2024 FOUNDRY MARK FLS									
PO NUMBER & DATE 10115103 Dt : 24-Nov-2023			ORDER ACCEPTANCE NO. & DATE SO-24-0377											
CHEMICAL COMPOSITION % wt														
MELTING PRACTICE : Steel melted using Electric Induction furnace, and fully killed using Aluminium.														
SPECIFICATION	C	Si	S	P	Mn	Ni	Cr	Mo	V	Cu	Al	--	*RE	**CE
Minimum	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum	0.23	0.60	0.020	0.025	1.20	0.50	0.50	0.20	0.030	0.30	0.08	--	1.00	0.43
Achieved	0.190	0.469	0.008	0.014	0.950	0.055	0.082	0.017	0.006	0.010	0.045	--	0.17	0.37
Heat Treatment : NORMALISING: Temperature raised to 920°C, Soaked for 3 hours and then atmospheric air cooled														
Visual Inspection - Castings conforms to ANSI/MSS-SP-55-2011 - Type II to XII (a)														
MECHANICAL PROPERTIES														
TEST TYPE	Tensile as per ASTM A370-22 / ISO 6892-1-2020 (at Room Temperature)						Impact (CVN) ISO 148-2016/ASTM A370-22		BEND TEST		HARDNESS			
SPECIFICATION	YS Rp 0.2% (KSI)	YS Rp 1.0% (KSI)	UTS (KSI)	% Elongation 4d 5d		% RA	Test Temperature (at °C) J Values Avg.		ANGLE	D	(HBW / HRC)			
Minimum	40	--	70	22	--	35	--	--	--	--	--			
Maximum	--	--	95	--	--	--	--	--	--	--	237			
Achieved	45.4	--	74.1	35.3	--	60.8	--	--	--	--	148			
DETAIL OF CASTINGS														
S.No.	PO S.No.	DESCRIPTION	CAST DRAWING NO / REV	PART NO		POURED QTY	DESP. QTY							
1	01	4" 900 FE BODY	D-487876 / 20	N2757928-60		5	1							
REMARKS: 1. Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast keel block. 2. *RE = Ni + Cr + Mo + V + Cu 3. ** CE = C + $\frac{Mn}{6}$ + $\frac{Cr+Mo+V}{5}$ + $\frac{Ni+Cu}{15}$ 4. The above stated KSI values have been calculated based on original tested N/mm ² values. 5. Stated actual Hardness value is less than 22 HRC. 6. Degassing process conform as per RMP 61496 Rev A.														
CERTIFIED THAT 1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO. 2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1. 3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIO ACTIVE ELEMENT OR RADIATION.							INSPECTED / REVIEWED & ACCEPTED / NOT ACCEPTED Sign & Date:  Sample Qty: M. VADIVEL - FICPI, QA  C. RAJAN ASSISTANT MANAGER - QUALITY MANUFACTURER'S AUTHORIZED INSPECTOR							
PREPARED BY: A.Yamuna 		CHECKED BY: N.KARTHICK  QUALITY ASSURANCE ENGINEER												

	FLOW LINK SYSTEMS PRIVATE LIMITED FOUNDRY UNIT-III-SF NO 395/2B2, KEMPANAICKANPALAYAM, ANNUR, COIMBATORE - 641 653, INDIA		INSPECTION CERTIFICATE (EN 10204:2004-3.1)		TC NO : 0781 DATE : 23-May-2024									
	CUSTOMER FLOWSERVE INDIA CONTROLS PVT LTD, B8, MMDA Industrial Area, Maraimalai Nagar-603209 Kancheepuram, INDIA		SPECIFICATION REFERENCE 1. ASTM A216-21 Gr.WCC 2. Customer Spec. RMC 90012 Rev. 15 3. NACE MR0175-2021 / ISO15156-2020		HEAT NO. 1N418 Date of Pouring 28-Mar-2024 FOUNDRY MARK FLS									
PO NUMBER & DATE 10115103 Dt : 24-Nov-2023			ORDER ACCEPTANCE NO. & DATE 24-0377											
CHEMICAL COMPOSITION % wt														
MELTING PRACTICE : Steel melted using Electric Induction furnace, and fully killed using Aluminium.														
SPECIFICATION	C	Si	S	P	Mn	Ni	Cr	Mo	V	Cu	Al	--	*RE	**CE
Minimum	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum	0.23	0.60	0.020	0.025	1.20	0.50	0.50	0.20	0.030	0.30	0.08	--	1.00	0.43
Achieved	0.203	0.432	0.009	0.015	1.016	0.025	0.025	0.009	0.007	0.010	0.039	--	0.08	0.38
Heat Treatment : NORMALISING: Temperature raised to 920°C, Soaked for 3 hours and then atmospheric air cooled														
Visual Inspection - Castings conforms to ANSI/MSS-SP-55-2011 - Type II to XII (a)														
MECHANICAL PROPERTIES														
TEST TYPE	Tensile as per ASTM A370-22 / ISO 6892-1-2020 (at Room Temperature)						Impact (CVN) ISO 148-2016/ASTM A370-22		BEND TEST		HARDNESS			
SPECIFICATION	YS Rp 0.2% (KSI)	YS Rp 1.0% (KSI)	UTS (KSI)	% Elongation 4d 5d		% RA	Test Temperature (at °C) J Values Avg.		ANGLE	D	(HBW / HRC)			
Minimum	40	--	70	22	--	35	--	--	--	--	--			
Maximum	--	--	95	--	--	--	--	--	--	--	237			
Achieved	45.7	--	76.2	32.2	--	56.3	--	--	--	--	152			
DETAIL OF CASTINGS														
S.No.	PO S.No.	DESCRIPTION	CAST DRAWING NO / REV		PART NO		POURED QTY		DESP. QTY					
1	01	4" 900 FE BODY	D-487876 / 20		N2757928-60		2		1					
REMARKS: 1. Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast keel block. 2. *RE = Ni + Cr + Mo + V + Cu 3. ** CE = C + $\frac{Mn}{6}$ + $\frac{Cr+Mo+V}{5}$ + $\frac{Ni+Cu}{15}$ 4. The above stated KSI values have been calculated based on original tested N/mm ² values. 5. Stated actual Hardness value is less than 22 HRC. 6. Degassing process conform as per RMP 61496 Rev A.														
CERTIFIED THAT 1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO. 2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1. 3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIO ACTIVE ELEMENT OR RADIATION.						INSPECTED / REVIEWED & ACCEPTED / NOT ACCEPTED Sign & Date:  Sample Qty: M. VAIDIVEL - FICPI, QA  C. RAJAN ASSISTANT MANAGER - QUALITY MANUFACTURER'S AUTHORIZED INSPECTOR								
PREPARED BY: A.Yamuna 		CHECKED BY: N.KARTHICK  QUALITY ASSURANCE ENGINEER												

	FLOW LINK SYSTEMS PRIVATE LIMITED FOUNDRY UNIT-III-SF NO 395/2B2, KEMPANAICKANPALAYAM, ANNUR, COIMBATORE - 641 653, INDIA	INSPECTION CERTIFICATE (EN 10204:2004-3.1)	TC NO : 0782 DATE : 23-May-2024											
	CUSTOMER FLOWSERVE INDIA CONTROLS PVT LTD, B8, MMDA Industrial Area, Maraimalai Nagar-603209 Kancheepuram, INDIA	SPECIFICATION REFERENCE 1. ASTM A216-21 Gr.WCC 2. Customer Spec. RMC 90012 Rev. 15 3. NACE MR0175-2021 / ISO15156-2020	HEAT NO. 1N420 Date of Pouring 28-Mar-2024 FOUNDRY MARK FLS											
PO NUMBER & DATE 10115103 Dt : 24-Nov-2023		ORDER ACCEPTANCE NO. & DATE 24-0377												
CHEMICAL COMPOSITION % wt														
MELTING PRACTICE : Steel melted using Electric Induction furnace, and fully killed using Aluminium.														
SPECIFICATION	C	Si	S	P	Mn	Ni	Cr	Mo	V	Cu	Al	--	*RE	**CE
Minimum	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum	0.23	0.60	0.020	0.025	1.20	0.50	0.50	0.20	0.030	0.30	0.08	--	1.00	0.43
Achieved	0.203	0.443	0.009	0.015	0.991	0.020	0.020	0.009	0.005	0.010	0.042	--	0.06	0.38
Heat Treatment : NORMALISING: Temperature raised to 920°C, Soaked for 3 hours and then atmospheric air cooled														
Visual Inspection - Castings conforms to ANSI/MSS-SP-55-2011 - Type II to XII (a)														
MECHANICAL PROPERTIES														
TEST TYPE	Tensile as per ASTM A370-22 / ISO 6892-1:2020 (at Room Temperature)						Impact (CVN) ISO 148-2016/ASTM A370-22		BEND TEST		HARDNESS			
SPECIFICATION	YS Rp 0.2% (KSI)	YS Rp 1.0% (KSI)	UTS (KSI)	% Elongation 4d 5d		% RA	Test Temperature (at °C) J Values Avg.		ANGLE	D	(HBW / HRC)			
Minimum	40	--	70	22	--	35	--	--	--	--	--			
Maximum	--	--	95	--	--	--	--	--	--	--	237			
Achieved	46.9	--	77.7	36.1	--	59.5	--	--	--	--	155			
DETAIL OF CASTINGS														
S.No.	PO S.No.	DESCRIPTION	CAST DRAWING NO / REV		PART NO		POURED QTY		DESP. QTY					
1	01	4" 900 FE BODY	D-487876 / 20		N2757928-60		2		2					
REMARKS: 1. Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast keel block. 2. *RE = Ni + Cr + Mo + V + Cu 3. ** CE = $C + \frac{Mn}{6} + \frac{Cr+Mo+V}{5} + \frac{Ni+Cu}{15}$ 4. The above stated KSI values have been calculated based on original tested N/mm ² values. 5. Stated actual Hardness value is less than 22 HRC. 6. Degassing process conform as per RMP 61496 Rev A.														
CERTIFIED THAT 1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO. 2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1. 3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIO ACTIVE ELEMENT OR RADIATION. INSPECTED / REVIEWED & ACCEPTED / NOT ACCEPTED Sign & Date:  Sample Qty: M. VADIVEL - FICPL QA  C. RAJAN ASSISTANT MANAGER - QUALITY MANUFACTURER'S AUTHORIZED INSPECTOR														
PREPARED BY: A. Yamuna  CHECKED BY: N. KARTHICK  QUALITY ASSURANCE ENGINEER														

 CASTLEGATE	CASTLEGATE INDUSTRIAL HOLDINGS PRIVATE LIMITED S.F.No.387/1, Devanampalayam(P.O), Pollachi(TK), Coimbatore, 642 120 Phone :7397161500 04259264166, Mail :info@castlegatellc.com											ISO 9001 : 2015 CERTIFIED COMPANY					
	INSPECTION CERTIFICATE According to EN10204 - 3.1																

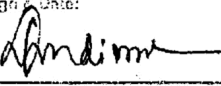
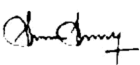
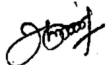
Customer : FLOWSERVE INDIA CONTROLS PVT LTD CHENNAI										T.C. No : FLS/4026 - 001 T.C.Date : 13/05/2024				
PO.No : 10116580 Dt. 20/02/2024										Heat No : 4026 Pour Date : 13/04/2024				
Specification : ASTM A216 GR.WCC (UNS J02503) - 2021										CASTING PROCESSED BY INDUCTION MELTING FURNACE				
										FOUNDRI IDENTIFICATION : CG				

CHEMICAL COMPOSITION %														
Element	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	CE	RE	AL	N
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.430	1.000	0.080	0.015
Actual	0.189	0.916	0.408	0.023	0.009	0.055	0.039	0.003	0.007	0.007	0.358	0.111	0.052	0.006

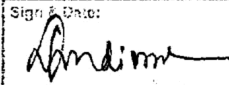
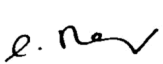
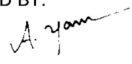
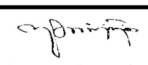
HEAT TREATMENT :	NORMALISING: Temperature raised to 920°C, soaked for 4 Hrs and then atmospheric air cooled.(Cycle No:C0018/24-25Dt.16/04/2024)													
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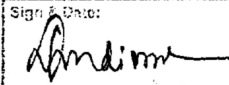
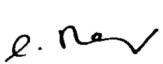
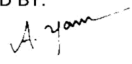
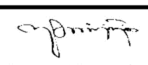
MECHANICAL PROPERTIES													
Requirements	Yield Strength		Tensile Strength		%Elongation	% Reduction	Hardness	Bend Test	Impact Test				
	kgf/mm ²	Mpa	kgf/mm ²	Mpa	GL :50 mm	In Area	HBW		CVN 2mm at MIN AVG 27 JULES				
	0.2 % Rp		-		-	-	-	-					
Minimum	28.03	275	49.44	485	22	35	---	---					
Maximum	---	---	66.77	655	---	---	235	---	1	2	3	Avg	
Actual	32.73	321.04	55.87	548.06	30.4	62.98	156	---	---	---	---	---	

ITEM DESCRIPTION				
SL.No.	Description	Drawing No.	Part No	Quantity
1	BODY CSTG 4" CL 900	00487876 REV 21	N487876WCC	4

Heat.SI.No :1,2,3,4 MP.No : MP19474,MP19475,MP19476,MP19477 ,				
Remarks: 1. This material conforms to NACE MR0175 - 2021 / ISO 15156 - 2 - 2020, FTDC001R7 & Material Spec No: RMC 90012 Rev15 & RMP 61158 Rev04. 2.Visual inspection as per MSS SP-55-2011 and found accepted. No surface irregularities to Type I none and Type II - XII not exceed 'a'. 3.Certified that the Castings are free from Radioactive contaminations. 4.MPI carried out as per MSS SP 53 and found accepted.				
INSPECTED / REVIEWED & ACCEPTED / NOT ACCEPTED Sign & Date:  M. VADIVEL - FICPL QA Sample Qty -				
We Confirm that the above material was manufactured,sampled,tested & inspected in accordance with the drawing,material specification,Customer order requirement and found to meet the requirements				
 Prepared By SABARI SANKAR		 Approved By ARUN KUMAR.N		

CIH/F/QA-04/R0

	FLOW LINK SYSTEMS PRIVATE LIMITED FOUNDRY UNIT-III-SF NO 395/2B2, KEMPANAICKANPALAYAM, ANNUR, COIMBATORE - 641 653, INDIA		INSPECTION CERTIFICATE (EN 10204:2004-3.1)		TC NO : 0816 DATE : 28-Jun-2024									
	CUSTOMER FLOWSERVE INDIA CONTROLS PVT LTD, B8, MMDA Industrial Area, Maraimalai Nagar-603209 Kancheepuram, INDIA		SPECIFICATION REFERENCE 1. ASTM A216-21 Gr.WCC 2. Customer Spec. RMC 90012 Rev. 15 3. NACE MR0175-2021 / ISO15156-2020		HEAT NO. 1N415 Date of Pouring 27-Mar-2024 FOUNDRY MARK FLS									
PO NUMBER & DATE 10115103 Dt : 24-Nov-2023			ORDER ACCEPTANCE NO. & DATE SO-24-0377											
CHEMICAL COMPOSITION % wt														
MELTING PRACTICE : Steel melted using Electric Induction furnace, and fully killed using Aluminium.														
SPECIFICATION	C	Si	S	P	Mn	Ni	Cr	Mo	V	Cu	Al	--	*RE	**CE
Minimum	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum	0.23	0.60	0.020	0.025	1.20	0.50	0.50	0.20	0.030	0.30	0.08	--	1.00	0.43
Achieved	0.194	0.466	0.009	0.015	0.979	0.058	0.087	0.017	0.005	0.011	0.041	--	0.18	0.38
Heat Treatment : NORMALISING: Temperature raised to 920°C, Soaked for 3 hours and then atmospheric air cooled														
Visual Inspection - Castings conforms to ANSI/MSS-SP-55-2011 - Type II to XII (a)														
MECHANICAL PROPERTIES														
TEST TYPE	Tensile as per ASTM A370-22 / ISO 6892-1-2020 (at Room Temperature)						Impact (CVN) ISO 148-2016/ASTM A370-22		BEND TEST		HARDNESS			
SPECIFICATION	YS Rp 0.2% (KSI)	YS Rp 1.0% (KSI)	UTS (KSI)	% Elongation 4d 5d		% RA	Test Temperature (at °C) J Values Avg.		ANGLE	D	(HBW / HRC)			
Minimum	40	--	70	22	--	35	--	--	--	--	--			
Maximum	--	--	95	--	--	--	--	--	--	--	237			
Achieved	45.8	--	77.0	36.0	--	60.0	--	--	--	--	154			
DETAIL OF CASTINGS														
S.No.	PO S.No.	DESCRIPTION	CAST DRAWING NO / REV		PART NO		POURED QTY		DESP. QTY					
1	01	4" 900 FE BODY	D-487876 / 20		N2757928-60		5		1					
REMARKS: 1. Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast keel block. 2. *RE = Ni + Cr + Mo + V + Cu 3. ** CE = C + $\frac{Mn}{6}$ + $\frac{Cr+Mo+V}{5}$ + $\frac{Ni+Cu}{15}$ 4. The above stated KSI values have been calculated based on original tested N/mm ² values. 5. Stated actual Hardness value is less than 22 HRC. 6. Degassing process conform as per RMP 61496 Rev A.														
CERTIFIED THAT 1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO. 2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1. 3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIO ACTIVE ELEMENT OR RADIATION.						INSPECTED / REVIEWED & ACCEPTED / NOT ACCEPTED Sign & Date:  Sample Qty: M. VADIVEL - FICPI, QA  C. RAJAN ASSISTANT MANAGER - QUALITY MANUFACTURER'S AUTHORIZED INSPECTOR								
PREPARED BY: A.Yamuna 		CHECKED BY: N.KARTHICK  QUALITY ASSURANCE ENGINEER												

	FLOW LINK SYSTEMS PRIVATE LIMITED FOUNDRY UNIT-III-SF NO 395/2B2, KEMPANAICKANPALAYAM, ANNUR, COIMBATORE - 641 653, INDIA		INSPECTION CERTIFICATE (EN 10204:2004-3.1)		TC NO : 0816 DATE : 28-Jun-2024									
	CUSTOMER FLOWSERVE INDIA CONTROLS PVT LTD, B8, MMDA Industrial Area, Maraimalai Nagar-603209 Kancheepuram, INDIA		SPECIFICATION REFERENCE 1. ASTM A216-21 Gr.WCC 2. Customer Spec. RMC 90012 Rev. 15 3. NACE MR0175-2021 / ISO15156-2020		HEAT NO. 1N415 Date of Pouring 27-Mar-2024 FOUNDRY MARK FLS									
PO NUMBER & DATE 10115103 Dt : 24-Nov-2023			ORDER ACCEPTANCE NO. & DATE SO-24-0377											
CHEMICAL COMPOSITION % wt														
MELTING PRACTICE : Steel melted using Electric Induction furnace, and fully killed using Aluminium.														
SPECIFICATION	C	Si	S	P	Mn	Ni	Cr	Mo	V	Cu	Al	--	*RE	**CE
Minimum	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum	0.23	0.60	0.020	0.025	1.20	0.50	0.50	0.20	0.030	0.30	0.08	--	1.00	0.43
Achieved	0.194	0.466	0.009	0.015	0.979	0.058	0.087	0.017	0.005	0.011	0.041	--	0.18	0.38
Heat Treatment : NORMALISING: Temperature raised to 920°C, Soaked for 3 hours and then atmospheric air cooled														
Visual Inspection - Castings conforms to ANSI/MSS-SP-55-2011 - Type II to XII (a)														
MECHANICAL PROPERTIES														
TEST TYPE	Tensile as per ASTM A370-22 / ISO 6892-1-2020 (at Room Temperature)						Impact (CVN) ISO 148-2016/ASTM A370-22		BEND TEST		HARDNESS			
SPECIFICATION	YS Rp 0.2% (KSI)	YS Rp 1.0% (KSI)	UTS (KSI)	% Elongation 4d 5d		% RA	Test Temperature (at °C) J Values Avg.		ANGLE	D	(HBW / HRC)			
Minimum	40	--	70	22	--	35	--	--	--	--	--			
Maximum	--	--	95	--	--	--	--	--	--	--	237			
Achieved	45.8	--	77.0	36.0	--	60.0	--	--	--	--	154			
DETAIL OF CASTINGS														
S.No.	PO S.No.	DESCRIPTION	CAST DRAWING NO / REV		PART NO		POURED QTY		DESP. QTY					
1	01	4" 900 FE BODY	D-487876 / 20		N2757928-60		5		1					
REMARKS: 1. Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast keel block. 2. *RE = Ni + Cr + Mo + V + Cu 3. ** CE = $C + \frac{Mn}{6} + \frac{Cr+Mo+V}{5} + \frac{Ni+Cu}{15}$ 4. The above stated KSI values have been calculated based on original tested N/mm ² values. 5. Stated actual Hardness value is less than 22 HRC. 6. Degassing process conform as per RMP 61496 Rev A.														
CERTIFIED THAT 1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO. 2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1. 3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIO ACTIVE ELEMENT OR RADIATION.						INSPECTED / REVIEWED & ACCEPTED / NOT ACCEPTED Sign & Date:  Sample Qty: M. VAIDIVEL - FICPI, QA  C. RAJAN ASSISTANT MANAGER - QUALITY MANUFACTURER'S AUTHORIZED INSPECTOR								
PREPARED BY: A.Yamuna 		CHECKED BY: N.KARTHICK  QUALITY ASSURANCE ENGINEER												

 CASTLEGATE	CASTLEGATE INDUSTRIAL HOLDINGS PRIVATE LIMITED S.F.No.387/1, Devanampalayam(P.O), Pollachi(TK), Coimbatore, 642 120 Phone :7397161500 04259264166, Mail :info@castlegatellc.com											ISO 9001 : 2015 CERTIFIED COMPANY					
	INSPECTION CERTIFICATE According to EN10204 - 3.1																

Customer : FLOWSERVE INDIA CONTROLS PVT LTD CHENNAI										T.C. No : FLS/4348 - 001 T.C.Date : 20/06/2024				
PO.No : 10118326 Dt. 27/05/2024										Heat No : 4348 Pour Date : 05/06/2024				
Specification : ASTM A216 GR.WCC (UNS J02503) - 2021										CASTING PROCESSED BY INDUCTION MELTING FURNACE				
										FOUNDRIY IDENTIFICATION : CG				

CHEMICAL COMPOSITION %														
Element	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	CE	RE	AL	N
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.430	1.000	0.080	0.015
Actual	0.201	0.980	0.453	0.021	0.011	0.072	0.033	0.007	0.008	0.009	0.385	0.129	0.042	0.010

HEAT TREATMENT :	NORMALISING: Temperature raised to 920°C, soaked for 6 Hrs and then atmospheric air cooled.(Cycle No:D0048/24-25Dt.08/06/2024)													
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MECHANICAL PROPERTIES														
Requirements	Yield Strength		Tensile Strength		%Elongation	% Reduction	Hardness	Bend Test	Impact Test					
	kgf/mm ²	Mpa	kgf/mm ²	Mpa	GL :50 mm	In Area	HBW		CVN 2mm at MIN AVG 27 JULES					
	0.2 % Rp		-		-	-	-	-						
Minimum	28.03	275	49.44	485	22	35	---	---						
Maximum	---	---	66.77	655	---	---	235	---	1	2	3	Avg		
Actual	30.27	296.91	51.62	506.42	28.2	48.09	149	---	---	---	---	---		

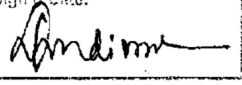
ITEM DESCRIPTION				
SL.No.	Description	Drawing No.	Part No	Quantity
1	BODY CSTG 4" CL 900	00487876 REV 21	N487876WCC	4

Heat.SI.No :1,2,3,4
MP.No : MP21279,MP21280,MP21281,MP21282 ,

Remarks:

1. This material conforms to NACE MR0175 - 2021 / ISO 15156 - 2 - 2020, FTDC001R7 & Material Spec No: RMC 90012 Rev15 & RMP 61158 Rev04.
2.Visual inspection as per MSS SP-55-2011 and found accepted. No surface irregularities to Type I none and Type II - XII not exceed 'a'.
3.Certified that the Castings are free from Radioactive contaminations.
4.MPI carried out as per ASME SECTION V Art.7 and found accepted.

INSPECTED / REVIEWED &
ACCEPTED / NOT ACCEPTED

Sign & Date:

M. VADIVEL - FICPL QA

Sample Qty:
-

We Confirm that the above material was manufactured,sampled,tested & inspected in accordance with the drawing,material specification,Customer order requirement and found to meet the requirements

 Prepared By SABARI SANKAR	 Approved By ARUN KUMAR.N
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CIH/F/QA-04/R0

 CASTLEGATE	CASTLEGATE INDUSTRIAL HOLDINGS PRIVATE LIMITED S.F.No.387/1, Devanampalayam(P.O), Pollachi(TK), Coimbatore, 642 120 Phone :7397161500 04259264166, Mail :info@castlegatellc.com											ISO 9001 : 2015 CERTIFIED COMPANY					
	INSPECTION CERTIFICATE According to EN10204 - 3.1																

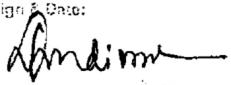
Customer : FLOWSERVE INDIA CONTROLS PVT LTD CHENNAI										T.C. No : FLS/4363 - 001 T.C.Date : 24/06/2024				
PO.No : 10118326 Dt. 27/05/2024										Heat No : 4363 Pour Date : 07/06/2024				
Specification : ASTM A216 GR.WCC (UNS J02503) - 2021										CASTING PROCESSED BY INDUCTION MELTING FURNACE				
										FOUNDRIY IDENTIFICATION : CG				

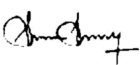
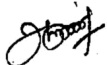
CHEMICAL COMPOSITION %														
Element	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	CE	RE	AL	N
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.430	1.000	0.080	0.015
Actual	0.206	0.906	0.421	0.023	0.012	0.133	0.074	0.016	0.013	0.008	0.394	0.244	0.041	0.009

HEAT TREATMENT :	NORMALISING: Temperature raised to 920°C, soaked for 4 Hrs and then atmospheric air cooled.(Cycle No:D0056/24-25Dt.13/06/2024)													
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MECHANICAL PROPERTIES														
Requirements	Yield Strength		Tensile Strength		%Elongation	% Reduction	Hardness	Bend Test	Impact Test					
	kgf/mm ²	Mpa	kgf/mm ²	Mpa	GL :50 mm	In Area	HBW		CVN 2mm at MIN AVG 27 JULES					
	0.2 % Rp		-		-	-	-	-						
Minimum	28.03	275	49.44	485	22	35	---	---						
Maximum	---	---	66.77	655	---	---	235	---	1	2	3	Avg		
Actual	32.57	319.54	54.76	537.23	34.76	65.12	163	---	---	---	---	---		

ITEM DESCRIPTION				
SL.No.	Description	Drawing No.	Part No	Quantity
1	BODY CSTG 4" CL 900	00487876 REV 21	N487876WCC	4

Heat.SI.No :1,2,3,4 MP.No : MP21370,MP21371,MP21372,MP21373 ,				
Remarks: 1. This material conforms to NACE MR0175 - 2021 / ISO 15156 - 2 - 2020, FTDC001R7 & Material Spec No: RMC 90012 Rev15 & RMP 61158 Rev04. 2.Visual inspection as per MSS SP-55-2011 and found accepted. No surface irregularities to Type I none and Type II - XII not exceed 'a'. 3.Certified that the Castings are free from Radioactive contaminations. 4.MPI carried out as per ASME SECTION V Art.7 and found accepted.				
INSPECTED / REVIEWED & ACCEPTED / NOT ACCEPTED Sign & Date:  M VADIVEL - FICPL QA Sample Qty: -				

We Confirm that the above material was manufactured,sampled,tested & inspected in accordance with the drawing,material specification,Customer order requirement and found to meet the requirements				
 Prepared By SABARI SANKAR		 Approved By ARUN KUMAR.N		

CIH/F/QA-04/R0